# AI-Powered Credit Risk Intelligence Platform

NeoStats AI Engineering Internship (Round 1) — Final Evaluation Presentation

Role: AI Engineering Candidate | Track: Machine Learning, XAI & Full-Stack AI

Dataset: Home Credit Default Risk (307,511 Applicants)

# Slide 2: Problem Statement

Key Challenges in Retail Banking Credit Underwriting

- **Severe Class Imbalance:** Retail loan defaults represent ~8% of applicants; naive classifiers predict all non-defaults, achieving 92% accuracy with 0% default detection.
- **Black-Box AI Resistance:** Regulatory standards (Basel III, Fair Lending, FCRA) require explainable, transparent reasons for credit risk classifications.
- **Disparate Relational Tables:** Critical borrower signals reside across fragmented databases (Credit Bureau, previous loans, repayment installments).
- **Data Accessibility Gap:** Credit officers and risk analysts need plain-English conversational access to live data without writing manual SQL queries.

# Slide 3: Business Objectives & Success Criteria

End-to-End Decision Support & Conversational Intelligence

- **Predictive Risk Scoring:** Accurately predict probability of default on a leak-free production pipeline.
- **Audit-Ready Explainability:** Unpack individual predictions using local TreeSHAP factor attributions.
- **Data-Driven Policy Rules:** Formulate 6 evidence-backed analytical underwriting rules derived from ML insights.
- **Conversational Data Assistant:** Deliver a secure NL-to-SQL interface grounded exclusively in database facts.
- **Full Containerization:** Package platform for one-command deployment via Docker.

# Slide 4: Dataset Architecture & Schema Strategy

Full Applicant Preservation with Leak-Free Aggregations

- **Master Anchor Table:** HC_application_train.csv (307,511 applicants, 122 raw features).
- **Target Distribution:** 282,686 Non-Defaults (91.93%) vs 24,825 Defaults (8.07%) — Imbalance Ratio 11.39 : 1.
- **Credit Bureau Aggregation:** Aggregated 1,716,428 bureau records into applicant-level features (loan count, active loans, max overdue days, total debt sum).
- **Previous Application Aggregation:** Aggregated 1,670,214 historical loan records (refusal rate, approval count, total credit granted).
- **Zero Data Loss:** Left join preserves 100% of the 307,511 applicants, rectifying the prior 96.4% inner-join data loss.

# Slide 5: Exploratory Data Analysis & Key Insights

Five Verified Empirical Findings from Historical Portfolios

- **1. External Credit Score Dominance:** Bottom decile of external scores (EXT_SOURCES) defaults at 21.4% vs 2.1% in top decile (10.2x spread).
- **2. Age Gradient:** Borrowers under 25 default at 12.3%, decreasing monotonically to 5.4% for borrowers aged 55+.
- **3. Debt Burden (DTI):** Default rates accelerate to 13.8% when requested credit exceeds 4.5x annual income vs 6.7% for DTI < 2.0x.
- **4. Educational Resilience:** Higher education holders default at 5.35% vs 10.93% for lower secondary education (2x differential).
- **5. Bureau Delinquency:** Borrowers with 30+ past overdue days in Credit Bureau suffer a 28.6% default rate vs 7.8% for clean records.

# Slide 6: Machine Learning Methodology

Strict Leak-Free Pipeline & Principled Imbalance Management

- **Data Partitioning:** 70% Train (215,257), 15% Validation (46,127), 15% Test (46,127) with stratified target sampling.
- **Strict Preprocessing Pipeline:** Median imputation for numerics, most_frequent imputation and OneHotEncoding for categoricals, fitted strictly on Train.
- **Class Imbalance Management:** scale_pos_weight=11.39 for gradient boosting models; class_weight="balanced" for linear/tree models.
- **Elimination of SMOTE Leakage:** Avoided pre-split oversampling that contaminated prior experimentation notebooks.

# Slide 7: Benchmark & Model Selection Evidence

Evaluated on Isolated Held-Out Test Split (46,127 records)

- LightGBM achieved the highest discriminatory power across both ROC-AUC (0.7716) and Precision-Recall AUC (0.2610).
- Model successfully identified 68.98% of all true loan defaults in the isolated test set.

| Algorithm | Test ROC-AUC | Test PR-AUC | Recall (Defaults) | Precision | F1-Score | Brier Score |
|---|---|---|---|---|---|---|
| LightGBM (Selected) | 0.7716 | 0.2610 | 68.98% | 17.72% | 0.2841 | 0.0670 (Calib) |
| XGBoost | 0.7695 | 0.2560 | 68.47% | 17.65% | 0.2831 | 0.1862 (Raw) |
| Logistic Regression | 0.7544 | 0.2342 | 68.29% | 16.52% | 0.2663 | 0.2037 (Raw) |
| Random Forest | 0.7507 | 0.2298 | 54.73% | 19.47% | 0.2872 | 0.0712 (Raw) |

# Slide 8: Probability Calibration & Empirical Risk Bands

Calibrated Posterior Probabilities with Defensible Threshold Selection

- **Probability Calibration:** Sigmoid calibration on validation data reduced Brier score loss by 63.9% (from 0.1856 to 0.0670).
- **Quantile-Based Thresholding:** Evaluated 20 validation quantile bins to define data-justified cutoffs:
- • **LOW RISK (< 2.93%):** Validated empirical default rate is strictly below 3.0% (Fast-track approval).
- • **MEDIUM RISK (2.93% to 15.31%):** Standard underwriting review band.
- • **HIGH RISK (>= 15.31%):** High-risk concentration band where empirical default rate exceeds 24.8% (Credit committee review).

# Slide 9: Explainable AI — TreeSHAP Feature Attribution

Transparent Local Explanations for Individual Loan Decisions

- **Individual Attribution:** TreeSHAP computes exact marginal contributions of each feature to the final prediction log-odds.
- **Risk-Increasing Drivers:** Clearly highlights factors pushing probability up (e.g., low external score, high credit burden, bureau overdue).
- **Risk-Reducing Drivers:** Highlights protective mitigants (e.g., strong bureau rating, stable employment, low debt-to-income).
- **Human-Readable Translation:** Automatically converts raw technical variable names into professional banking terminology.

# Slide 10: Analytical Business Rules Engine

Data-Derived Rules Bridging Machine Learning & Underwriting Guidelines

- **Compliance Designation:** Strictly labelled as analytical model-derived decision aids, not official statutory banking policy.
- **Rule 1:** Severe External Bureau Score Depletion (EXT_SOURCES_MEAN < 0.30 -> High Risk Amplifier).
- **Rule 2:** Excessive Debt-to-Income Leverage (Credit / Income > 4.5x or Annuity / Income > 35%).
- **Rule 3:** Historical Bureau Delinquency (Max days overdue > 30 days or overdue debt > $5,000).
- **Rule 4:** Institutional Rejection Precedent (Past loan refusal rate >= 50%).
- **Rule 5:** Youth & Short Employment Tenancy (Age < 26 and employed < 1.5 years).
- **Rule 6 (Protective):** Prime Stability Profile (EXT_SOURCES_MEAN > 0.65 and clean bureau history).

# Slide 11: Conversational Talk-to-Data Architecture

Natural Language Question -> Validated SQL -> Database Result -> Plain-English Answer

- **Grounded SQLite Warehouse:** credit_risk_warehouse.db indexed on applicants, bureau, and previous applications.
- **Dual Generation Modes:** Supports external LLMs (Gemini / OpenAI) + Offline Deterministic Pattern Engine for zero-dependency operation.
- **Immutable Read-Only Execution:** Queries execute under read-only URI connection mode (mode=ro).
- **Zero Hallucinated Numbers:** All numeric metrics in chat answers are extracted directly from database execution tables.

# Slide 12: SQL Security Validation & Hallucination Guardrails

Strict Pre-Execution Protection and Unsupported Query Interception

- **Pre-Execution AST Validation:** sql_validator.py runs strictly BEFORE query execution.
- **Destructive Command Blocking:** Rejects INSERT, UPDATE, DELETE, DROP, ALTER, CREATE, ATTACH, PRAGMA.
- **Multiple Statement Blocking:** Rejects semicolon chaining and malicious injection vectors.
- **Hallucination Guardrails:** Intercepts queries requesting unsupported attributes (e.g. eye colour, pet, vehicle brand) and returns transparent refusal messages.

# Slide 13: Interactive Streamlit UI Platform

Modular, User-Friendly Multi-Section Interface

- **1. Executive Overview:** Portfolio KPI cards, benchmark comparison table, and threshold methodology.
- **2. Exploratory Data Analysis:** 5 interactive Plotly visualizations with clear business interpretations.
- **3. Risk Scoring Form:** User-friendly applicant parameter inputs with default profiles and risk band badges.
- **4. SHAP Waterfall Visualization:** Interactive waterfall plot decomposing local risk drivers.
- **5. Business Rules Explorer:** Live rule evaluator tracking active policy alerts for the current applicant.
- **6. Talk-to-Data Chatbot:** Interactive conversational assistant with transparent SQL code inspection.

# Slide 14: Modular Production Codebase Structure

Clean Engineering Separation of Concerns

- src/data/: Memory-optimized data loader and applicant-level aggregation modules.
- src/preprocessing/: Leak-free ColumnTransformer pipeline.
- src/models/: Training, evaluation, calibration, and inference engines.
- src/explainability/: TreeSHAP explainer with human-readable feature mappings.
- src/rules/: Analytical business rules engine.
- src/chatbot/: Security validator, prompt templates, query executor, and response formatter.
- tests/: Automated unit and integration test suite.
- app/: Streamlit multi-tab web application.

# Slide 15: Containerization & Deployment Architecture

Reproducible One-Command Deployment via Docker

- **Dockerfile:** Python 3.11-slim base image with compiled OpenMP (libgomp1) support for LightGBM and XGBoost.
- **docker-compose.yml:** Orchestrates web application container with environment configuration.
- **Healthcheck:** Integrated Streamlit healthcheck probe for production monitoring.
- **Execution:** Evaluator can run docker compose up to launch the complete platform on port 8501.

# Slide 16: Conclusion & Business Value

Summary of Internship Deliverables and Platform Capabilities

- **End-to-End Delivery:** Built complete, working, explainable AI platform without deleting or overwriting experimentation research.
- **Technical Integrity:** Fixed prior row-loss and data leakage issues; benchmarked 4 models on full 307k applicant dataset.
- **Explainability & Safety:** Paired state-of-the-art LightGBM performance (0.7716 ROC-AUC) with SHAP attribution and SQL injection defenses.
- **Grounded Intelligence:** Delivered reliable Conversational Talk-to-Data capability where database facts remain the single source of truth.

# Slide 17: Future Improvements & Next Steps

Path to Enterprise Production Scaling

- **Dynamic Credit Limit Optimization:** Extend from binary default risk to continuous expected loss (EL = PD x LGD x EAD).
- **Graph Neural Networks (GNN):** Incorporate social circle graph connections (DEF_30 / DEF_60 social circles) for fraud ring detection.
- **Automated Model Monitoring:** Implement Population Stability Index (PSI) and concept drift alerts on live applicant distributions.
- **Multi-Turn Query Optimization:** Enhance SQL dialect translation to support complex cross-table recursive joins.